MIT-WPU
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MIT WORLD PEACE UNIVERSITY | PUNE
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

PRN:

End Semester Examination

May-June 2023

EEE2005B - Data Structures and Algorithms

Schedule ID: 18377

Faculty/School	Faculty of Engineering & Technology	**Term**	IV
Program	Second Year B. Tech	**Duration**	**1 Hours 30 Minutes**
Specialization		**Max. Marks**	**40**

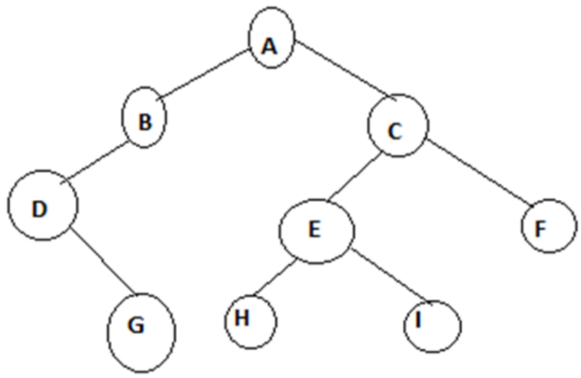
Read the instructions provided for every question properly before attempting the answer.

Section - 1 : contain(s) **10** question(s) and each question carries **5** mark(s).You can answer any **8** questions out of **10.**

Click **Finish** only after completion of the Exam.

Section - 1 (8 X 5 Marks)
Answer any 8 questions

1	What is Abstract Datatype(ADT) and explain ADT with example.	**5 marks**	CO1	Remembering
2	Give the formal definitions of O,Ω and ϴ notations with proper graphs.	**5 marks**	CO1	Understanding
3	List the functions used in dynamic memory allocation and brief about malloc() function.	**5 marks**	CO2	Remembering
4	List all and explain any two applications of stack in detail with example.	**5 marks**	CO2	Remembering
5	Convert the given infix expression **A*(B+C)-D/E** to postfix expression using stack.(Steps carries marks).	**5 marks**	CO3	Analysing
6	Explain Job Scheduling with block diagram.	**5 marks**	CO2, CO3	Remembering

7	Perform inorder, preorder and postorder traversals on the given tree . 	**5 marks**	CO3	Remembering
8	List all and explain any 4 tree terminologies with example for each.	**5 marks**	CO3	Remembering
9	List all and explain any 5 graph terminologies with example.	**5 marks**	CO2, CO3	Remembering
10	Explain Dijkstra's Algorithm using example.(Note: Steps carries marks).	**5 marks**	CO3	Remembering

END OF QUESTION PAPER